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Describe how viruses are replicated.

[4 marks]

Question 3 continues on the next page**Turn over ►**

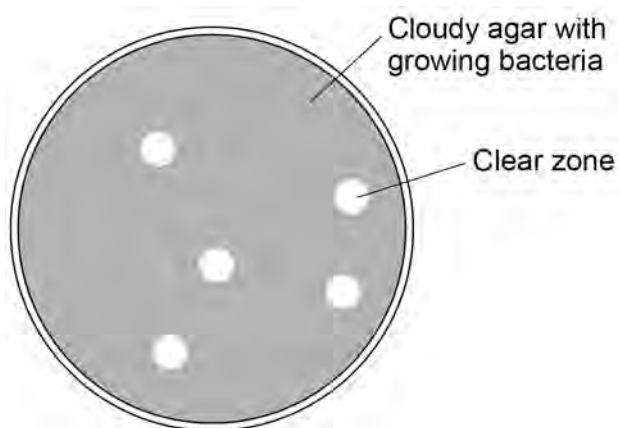
0 3 . 2 A scientist investigated the effect of a virus species on a bacterial culture.

The scientist:

- mixed a diluted sample of the virus with a bacterial culture in liquid agar
- poured the liquid agar into a Petri dish and let the agar set
- incubated the virus-bacterial culture at 30 °C for 24 hours.

Figure 3 shows the appearance of the Petri dish after 24 hours.

Figure 3



Explain why clear zones are present in **Figure 3**.

[1 mark]

0 3 . 3 Predict the appearance of the Petri dish if the incubation was extended by a few more hours.

[1 mark]



0 3 . 4

Animal cells replicate during the cell cycle.

Figure 4 shows one animal cell at a stage of the cell cycle.

Figure 4



Identify the stage of the cell cycle shown in **Figure 4**.

Explain the appearance of the cell shown in **Figure 4**.

Do **not** refer to organelles in your answer.

[2 marks]

Identity of stage _____

Explanation _____

8

Turn over for the next question

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